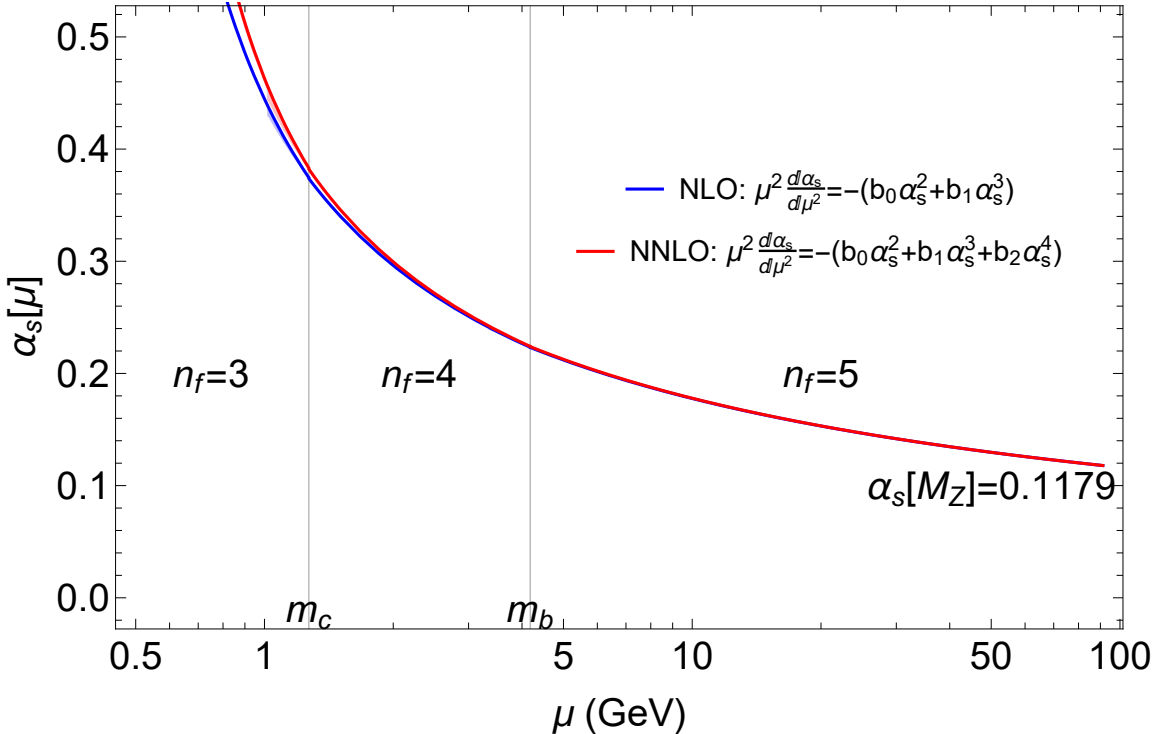
0.5
0.4
0.3
0.2
0.1
0.0
$\alpha_s[\mu]$
NLO: $\mu^2 \frac{d\alpha_s}{d\mu^2} = -(b_0\alpha_s^2 + b_1\alpha_s^3)$
NNLO: $\mu^2 \frac{d\alpha_s}{d\mu^2} = -(b_0\alpha_s^2 + b_1\alpha_s^3 + b_2\alpha_s^4)$
n_f=3
n_f=4
n_f=5
$\alpha_s[M_Z]$=0.1179
m_c
m_b
0.5
1
5
10
50
100
μ (GeV)